UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

*5503432532*

**PHYSICS** **5054/42**

Paper 4 Alternative to Practical **May/June 2010**

**1 hour**

Candidates answer on the Question Paper.

No Additional Materials are required.

**READ THESE INSTRUCTIONS FIRST**

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

This document consists of **9** printed pages and **3** blank pages.

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UNIVERSITY of CAMBRIDGE International Examinations

[Turn over

**1** A group of students performs an experiment to investigate the flow of oil at different temperatures.

200 $cm^3$ of oil is heated gently and its temperature $\theta$ is recorded. The oil is then poured through a funnel into a second beaker, as shown in Fig. 1.1.

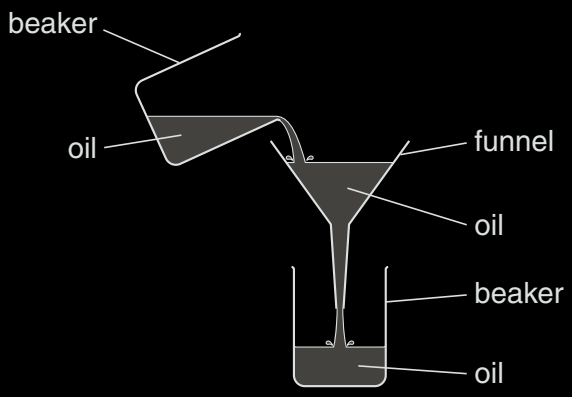



**Fig. 1.1**

The time $t$ taken for the oil to flow through the funnel is recorded with a stopwatch.

**(a)** State why it is important to stir the oil during heating.

.....

..... [1]

**(b)** Explain why the oil is heated **gently**.

.....

..... [1]

**(c)** The experiment is repeated for several values of $\theta$. Values of $\theta$ and $t$ are recorded in the table of Fig. 1.2.

| $\theta$/°C | $t$/s |
|---|---|
| 69 | 13.69 |
| 52 | 14.97 |
| 35 | 17.34 |
| 24 | 21.16 |
| 17 | 25.16 |
| 10 | 28.85 |

**Fig. 1.2**

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**(i)** On Fig. 1.3, plot the graph of $t$/s on the $y$-axis against $\theta$/°C on the $x$-axis.

Start your graph from $\theta = 0$ °C and $t = 12$ s. Draw the curved line of best fit.




**Fig. 1.3** [4]

**(ii)** Estimate the time taken for oil at 80 °C to flow through the funnel.

..... [1]

**(iii)** State the maximum reading on a standard laboratory liquid-in-glass thermometer that is suitable for this experiment.

..... [1]

**(d)** Explain why it is not possible to repeat a reading immediately after it is taken.

.....

..... [1]

**(e)** Tick **two** boxes to show which of the following will make the experiment more accurate.

| | |
|---|---|
| using a more sensitive thermometer | |
| using two people to take the measurements | |
| using a thicker oil | |
| using a larger range of readings | |
| pouring the oil quickly after taking its temperature | |

[2]

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**2** Fig. 2.1 shows a wooden metre rule with small holes drilled through it.

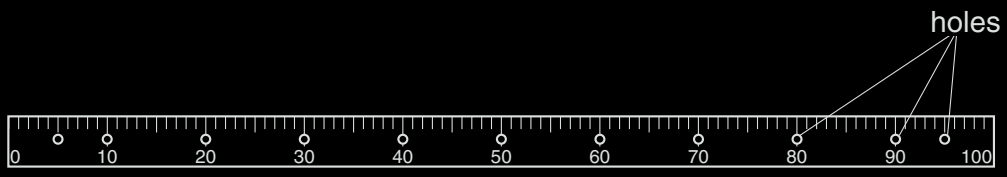



**Fig. 2.1**

The metre rule is suspended from the hole at 5.0 cm so that it can swing freely, as shown in Fig. 2.2.

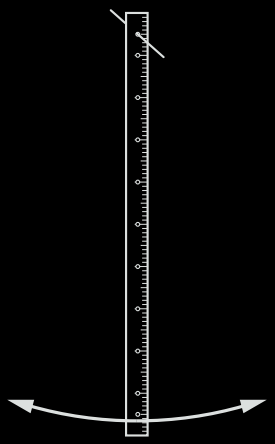


**Fig. 2.2**

The rule is made to swing from side-to-side and the time $T$ for one complete swing is determined.

**(a)** Describe three experimental techniques used to obtain an accurate value for $T$.

1. .....

.....

2. .....

.....

3. .....

.....

[3]

**(b)** The distance $d$ between a hole and the zero end of the rule is varied, by suspending the rule from different holes.

The time $T$ is determined for each value of $d$.

The results are recorded in the table of Fig. 2.3.

| $d$/cm | $T$/s |
|---|---|
| 5.0 | 1.61 |
| 10.0 | 1.57 |
| 20.0 | 1.52 |
| 30.0 | 1.58 |
| 40.0 | 1.91 |

**Fig. 2.3**

**(i)** Describe how $T$ varies with $d$.

.....

.....

.....

..... [2]

**(ii)** Suggest the value of $T$ for $d = 95.0$ cm.

$T =$ ..... [1]

**(iii)** Suggest why $T$ was not measured for $d = 50.0$ cm.

.....

.....

..... [1]

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**3** The efficiency of a motor is measured using the apparatus shown in Fig. 3.1.

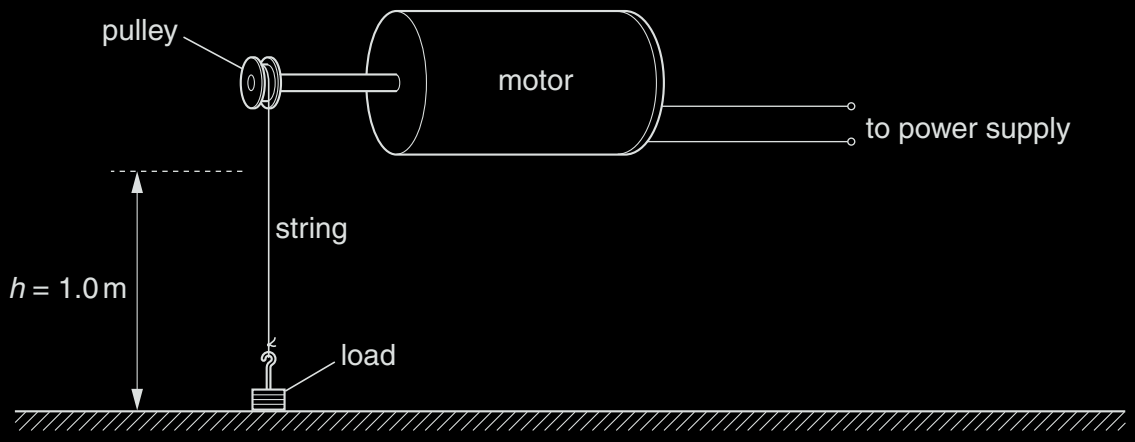



**Fig. 3.1** (not to scale)

When the motor is switched on, the string winds round the pulley and the load is lifted. Six students measure the time $t$ taken for the load to be lifted a height $h$ of 1.0 m.

**(a)** The times recorded are

4.47 s, 4.53 s, 4.39 s, 4.44 s, 5.92 s, 4.61 s.

**(i)** The result of 5.92 s is discarded.

Suggest why this result is very much larger than the others.

..... [1]

**(ii)** Calculate the average time $t_{av}$ of the other results.

Considering the variation in the students' results, give your answer to an appropriate number of significant figures. Show your working.

$t_{av}$ = ..... s [2]

**(b)** The efficiency $E$ of the motor is given by the relationship

$$E = \frac{2 \times 10^2}{3 \times t_{av}} \%.$$

Use this relationship to calculate $E$.

$E$ = ..... [1]

**(c)** Describe one way in which the students can measure $h$ accurately.

.....

.....

..... [1]

**4** A student performs an experiment using optical pins to find the effect of a circular block, made of transparent plastic, on parallel rays of light.

Fig. 4.1 shows the circular plastic block on a sheet of white paper.

The student draws round the block with a sharp pencil. The student also draws three parallel lines up to the block before starting the experiment.

**(a)** Explain why the student draws round the block.

.....

..... [1]

**(b)** The student places pins $P_1$ and $P_2$ on line 1, representing an incident ray of light.

Describe how the student places pins $P_3$ and $P_4$ to locate the emergent ray.

.....

.....

..... [1]

**(c)** On Fig. 4.1,

**(i)** complete the path of the ray along line 1 through the block, [1]

**(ii)** draw the normal at the point where the ray along line 1 enters the block, [1]

**(iii)** measure the angle of incidence *i* where the ray along line 1 enters the block.

*i* = ..... [1]

**(d)** Explain why a ray along line 2 passes through the block without changing direction.

.....

..... [1]

**(e)** On Fig. 4.1, complete the path of a ray along line 3 through the block. [1]

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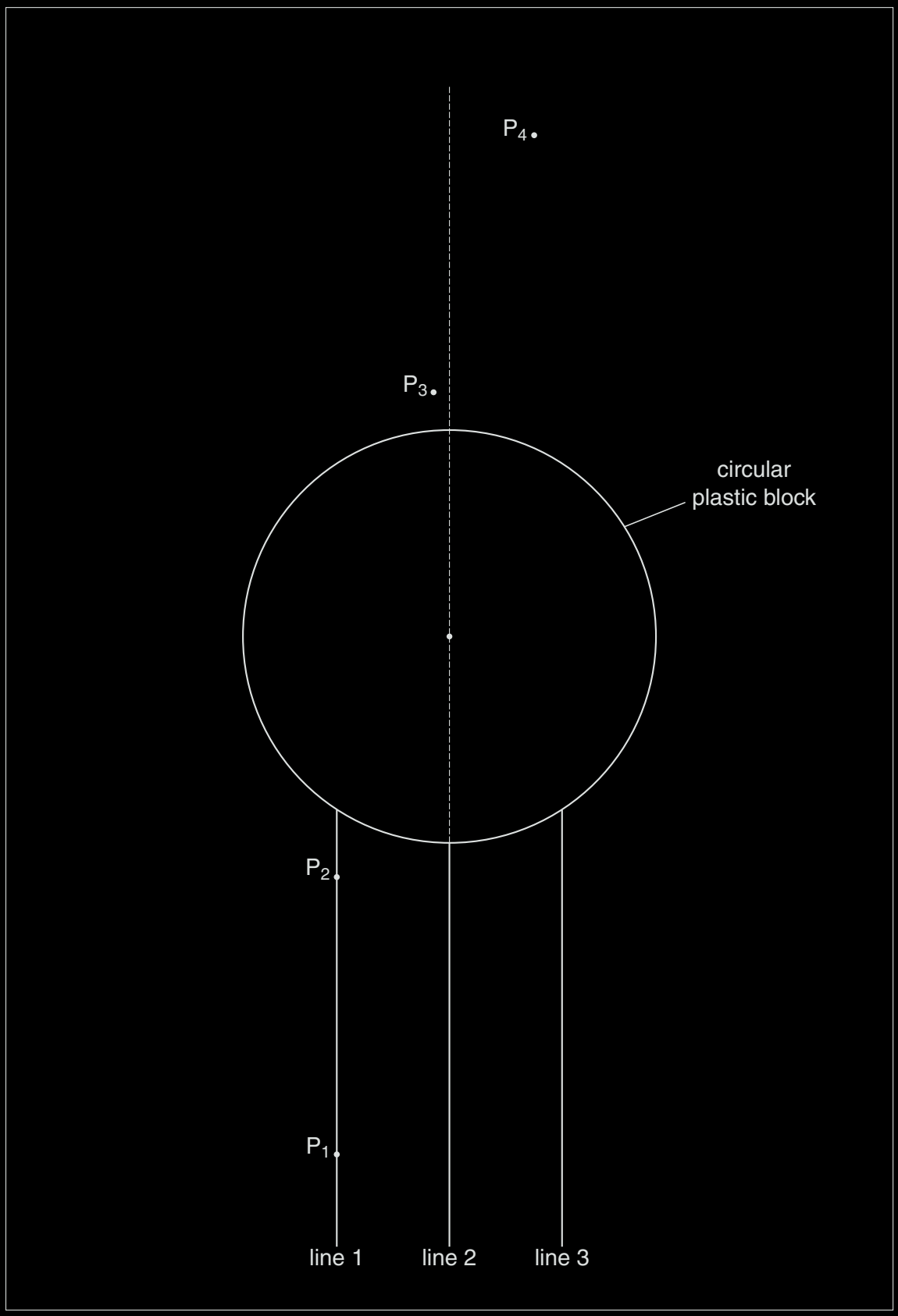



**Fig. 4.1**

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